

Natural optical activity with SOC at transparent frequencies

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Natural optical activity

Natural optical activity (NOA) refers to the first-order ***spatial dispersion*** of the macroscopic dielectric tensor

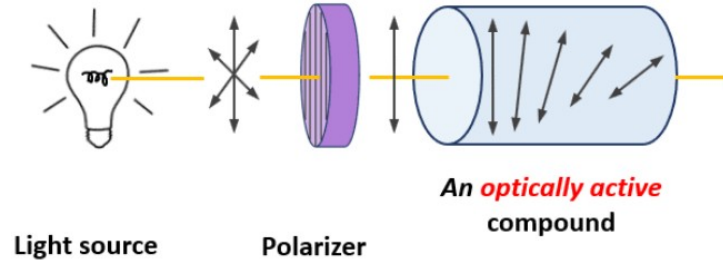
$$\epsilon_{\alpha\beta}(\omega, \mathbf{q}) = \epsilon_{\alpha\beta}(\omega, \mathbf{q} = \mathbf{0}) + iq_{\gamma}\eta_{\alpha\beta\gamma}(\omega) + \dots$$

Dielectric tensor

Frequency

Natural optical activity

TR-symmetric
insulators



Optical rotation: quantifies the rotation of the plane of polarization of linearly polarized light

$$\rho(\omega) \propto \omega^2 \eta_{\alpha\beta\gamma}(\omega)$$

Spatial dispersion in crystals with periodic boundary conditions?

Spatial dispersion effects in Abinit

Long-wavelength DFPT

M. Royo and M. Stengel, Phys. Rev. X **9**, 021050 (2019)

Flexoelectricity

Dynamical quadrupoles

Natural optical activity

A. Zabalo and M. Stengel, Phys. Rev. Lett. **131**, 086902 (2023)



$$\epsilon_{\alpha\beta}(\mathbf{q}) = \delta_{\alpha\beta} - \frac{4\pi}{\Omega} E_{\mathbf{q}}^{\epsilon_{\alpha}\epsilon_{\beta}} \xrightarrow{\text{analytical derivative in } q} \eta_{\alpha\beta\gamma} = -\frac{4\pi}{\Omega} \text{Im} E_{\gamma}^{\epsilon_{\alpha}\epsilon_{\beta}}$$

unit cell volume

$$\bar{\rho} \equiv \lim_{\omega \rightarrow 0} \rho(\omega) / (\hbar\omega)^2$$

Well-defined zero frequency limit

2n+1 theorem: first-order spatial dispersion coefficients can be computed with the knowledge of uniform field perturbations ($q=0$)

Current implementation

$$E_{\gamma}^{\varepsilon_{\alpha}\varepsilon_{\beta}} = \int_{\Omega} \int n^{\varepsilon_{\alpha}}(\mathbf{r}) \left. \frac{\partial}{\partial q_{\gamma}} K_{\text{Hxc}}(\mathbf{r}, \mathbf{r}') \right|_{\mathbf{q}=0} n^{\varepsilon_{\beta}}(\mathbf{r}') d^3r d^3r' + 2s \int_{\text{BZ}} \Omega \frac{d^3k}{(2\pi)^3} E_{\mathbf{k},\gamma}^{\varepsilon_{\alpha}\varepsilon_{\beta}}$$

First-order electron densities

Gradient in q of the Hartree and exchange-correlation kernel

Spin multiplicity

Required ingredients

- Ground state wave-functions: $|u_{m\mathbf{k}}^{(0)}\rangle$
- 3x d/dk wave-functions: $|\partial_{\alpha} u_{m\mathbf{k}}^{(0)}\rangle \equiv |u_{m\mathbf{k}}^{k_{\alpha}}\rangle$
- 3x d/dE wave-functions: $|u_{m\mathbf{k}}^{\varepsilon_{\beta}}\rangle$
- 3x d/dB wave-functions (orbital part only, no SCF): $|u_{m\mathbf{k}}^{B_{\delta}}\rangle$
- 6x $d^2/dkdk$ wave-functions: $|\partial_{\alpha\beta}^2 u_{m\mathbf{k}}^{(0)}\rangle$

New methodological developments

- 1) New analytical expression (remove $d^2/dkdk$)
- 2) Include SOC
- 3) Frequency dependent NOA

1) New analytical expression

New analytical expression

$$\int_{\text{BZ}} d^3k \left[E_{\mathbf{k},\gamma}^{\varepsilon_\alpha \varepsilon_\beta} - f(\mathbf{k}) \right] \longrightarrow \int_{\text{BZ}} d^3k f(\mathbf{k}) = 0$$

Our new expression doesn't require $d^2/dkdk$ wave-functions!

$$f(\mathbf{k}) = \frac{1}{2} \frac{dE_{\mathbf{k},\mathbf{q}}^{\varepsilon_\alpha \varepsilon_\beta}}{dk_\gamma} \Big|_{\mathbf{q}=0}$$

Useful because

- i) Remove 6 response function calculations
- ii) The $d^2/dkdk$ terms are not yet implemented with SOC

2) Including SOC

New implementation

- Ground state wave-functions: $|u_{m\mathbf{k}}^{(0)}\rangle$
- 3x d/dk wave-functions: $|\partial_\alpha u_{m\mathbf{k}}^{(0)}\rangle \equiv |u_{m\mathbf{k}}^{k_\alpha}\rangle$
- 3x d/dE wave-functions: $|u_{m\mathbf{k}}^{\mathcal{E}_\beta}\rangle$
- 3x d/dB wave-functions (**spin+orbital part, with SCF**): $|u_{m\mathbf{k}}^{\mathcal{B}_\delta}\rangle$
- ~~· 6x d²/dkdk wave-functions: $|\partial_{\alpha\beta}^2 u_{m\mathbf{k}}^{(0)}\rangle$~~



It is sufficient to include SOC in the individual contributions to the NOA

B-field response requires new theoretical and computational developments

Including SOC

SOC at the level of wave functions only

$$|\psi_{m\mathbf{k}}\rangle \longrightarrow \begin{pmatrix} |\psi_{m\mathbf{k}}^\alpha\rangle \\ |\psi_{m\mathbf{k}}^\beta\rangle \end{pmatrix}$$

α and β are spin components

Pauli vector, containing the Pauli matrices

$$\boldsymbol{\sigma} = (\sigma^x, \sigma^y, \sigma^z)$$

m : magnetization density

SOC at the level of wave functions *and densities*

$$n_{\alpha\beta}^{(0)} = \boxed{\frac{1}{2}n_0\delta_{\alpha\beta}} + \boxed{\frac{1}{2}\mathbf{m} \cdot \boldsymbol{\sigma}_{\alpha\beta}} \longrightarrow n_{\alpha\beta}^\lambda = \frac{1}{2}n_0^\lambda\delta_{\alpha\beta} + \boxed{\frac{1}{2}\mathbf{m}^\lambda \cdot \boldsymbol{\sigma}_{\alpha\beta}}$$

Charge density

Vanishes in TR-symmetric (nonmagnetic) insulators

Nonzero if λ breaks TR!

(B-field response)

B-field response

$$|u_{m\mathbf{k}}^{\mathcal{B}_c}\rangle = \underbrace{-i \varepsilon_{abc} \partial_a \hat{P}_{\mathbf{k}} \partial_b \hat{P}_{\mathbf{k}}}_{\text{valence-band (orb)}} |u_{m\mathbf{k}}^{(0)}\rangle + \underbrace{| \bar{u}_{m\mathbf{k}}^{\mathcal{B}_c} \rangle}_{\text{conduction-band (spin+orb)}}$$

$$\hat{P}_{\mathbf{k}} = \sum_m |u_{m\mathbf{k}}^{(0)}\rangle \langle u_{m\mathbf{k}}^{(0)}|, \quad \hat{Q}_{\mathbf{k}} = \mathbb{I} - \hat{P}_{\mathbf{k}}$$

$$\partial_a \hat{P}_{\mathbf{k}} = \sum_m \left(|\partial_a u_{m\mathbf{k}}^{(0)}\rangle \langle u_{m\mathbf{k}}^{(0)}| + |u_{m\mathbf{k}}^{(0)}\rangle \langle \partial_a u_{m\mathbf{k}}^{(0)}| \right)$$

The linear-response part fulfills the following Sternheimer equation

$$\hat{Q}_{\mathbf{k}} \left(\hat{H}_{\mathbf{k}}^{(0)} - \epsilon_{m\mathbf{k}}^{(0)} \right) \hat{Q}_{\mathbf{k}} | \bar{u}_{m\mathbf{k}}^{\mathcal{B}_c} \rangle = -\hat{Q}_{\mathbf{k}} \left(\underbrace{-\frac{i}{2} \varepsilon_{abc} \{ \partial_a \hat{P}_{\mathbf{k}}, \partial_b \hat{H}_{\mathbf{k}}^{(0)} \}}_{\text{orbital}} - \underbrace{\frac{1}{2} \hat{\sigma}_c}_{\text{spin}} + \underbrace{\hat{V}^{\mathcal{B}_c}}_{\text{SCF}} \right) |u_{m\mathbf{k}}^{(0)}\rangle$$

$\hat{V}^{\mathcal{B}_c} [n^{\mathcal{B}_c}]$

Including SOC involves



spin B-field contributions

$$1) \quad \mathbf{B} \text{ couples to spins } \hat{H}^{B_z^{\text{spin}}} = -\frac{1}{2} \hat{\sigma}^z \longrightarrow V_{\text{SCF}}^{B^{\text{spin}}}(\mathbf{r})$$

$$2) \quad n^{B^{\text{orb}}}(\mathbf{r}) \neq 0 \longrightarrow V_{\text{SCF}}^{B^{\text{orb}}}(\mathbf{r}) \text{ (new)}$$

$$n_{\alpha\beta}^{B^{\text{orb}}} = \frac{1}{2} n_0^{B^{\text{orb}}} + \frac{1}{2} \mathbf{m}^{B^{\text{orb}}} \cdot \boldsymbol{\sigma}_{\alpha\beta}$$

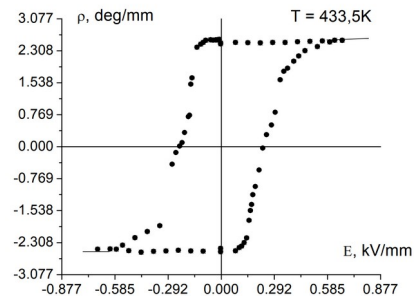
$\nearrow 0$

$P\bar{6}$ $\longrightarrow$ $P3$ *Ferroelectric, chiral and optically active!*
Paralecetric

No theoretical studies about PGO's NOA!
Unusually large SOC effects on PGO's energy landscape

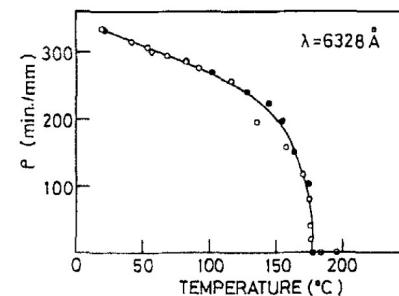
M. Fava *et al.*, Phys. Rev. B **109**, 024113 (2024)
 M. Fava *et al.*, Phys. Rev. B **108**, L201112 (2023)

Gyroelectric effect



D. Adamenko, *et al.*, Ukr. J. Phys. Opt. 8, 42 (2007)

Optical rotation vs T

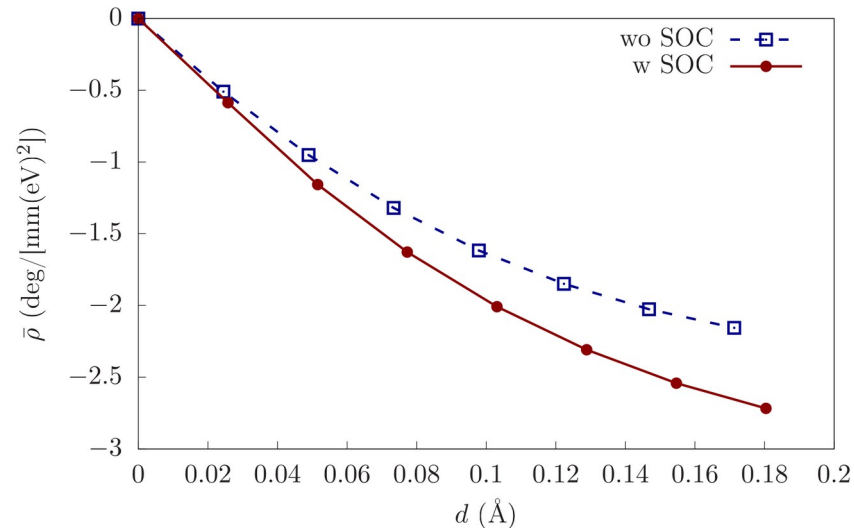
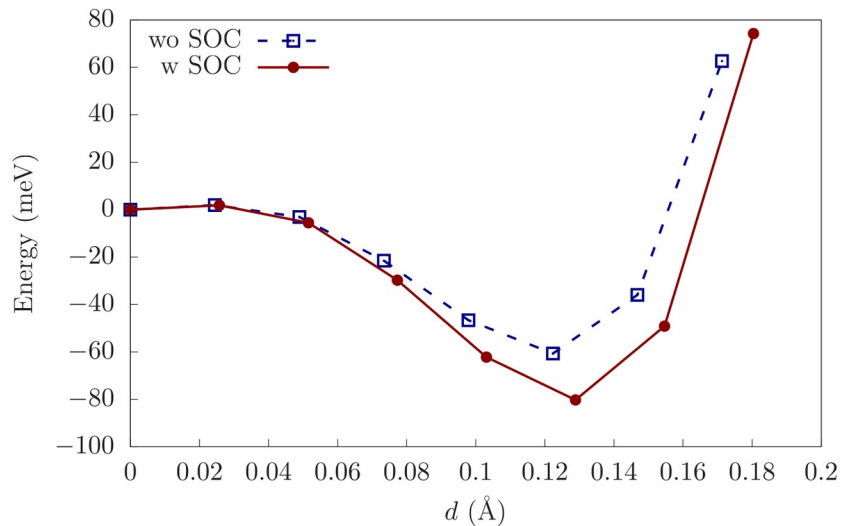


H. Iwasaki, *et al.*, Ferroelectrics 3, 157 (1972)

Effects of SOC on NOA

- 1) NOA as a function of the FE distortion, $P\bar{6} \longrightarrow P3$
- 2) Spin **B**-field contributions

Test case: $\text{Pb}_5\text{Ge}_3\text{O}_{11}$



Linear interpolation at fixed cell parameters: $P\bar{6} \rightarrow P3$

$$d = \sqrt{\frac{1}{M} \sum_{\kappa} m_{\kappa} |\mathbf{u}_{\kappa}|^2}$$

$\mathbf{u}_{\kappa}$: displacement from PE to FE phase

m_{κ} : mass of ion κ

M : unit cell mass

Spin B-field contributions

	no SOC	SOC wo spins	SOC w spins	Exp.
$\bar{\rho}$ (deg/[mm(eV) ²])	−1.95	−2.42	−2.48	−2.6 ^a , −2.3 ^b

SOC w spins: SOC included, full treatment of spins

SOC wo spins: SOC included, no spin **B**-field contributions

^a H. Iwasaki *et al.*, J. Appl. Phys. 43, 4907 (1972)

^b D. Adamenko *et al.*, Condens. Matter Phys. 25, 43703 (2022)

Impact of SOC: Strong SOC effects on PGO's NOA

Spin-B contributions: Further amplify the rest of SOC effects. Will be more significant in magnetic materials

3) Frequency dependent NOA

Frequency dependent NOA

Frequency dependent NOA $\longrightarrow$ All we need is the response to a dynamical electric field within DFPT

$$\hat{Q}_{\mathbf{k}} \left(\hat{H}_{\mathbf{k}}^{(0)} - \epsilon_{m\mathbf{k}}^{(0)} - \omega \right) \hat{Q}_{\mathbf{k}} |u_{m\mathbf{k}}^{\mathcal{E}_{\beta}}(\omega)\rangle = -\hat{Q}_{\mathbf{k}} \left[\hat{H}_{\mathbf{k}}^{\mathcal{E}_{\beta}} + \hat{V}^{\mathcal{E}_{\beta}}(\omega) \right] |u_{m\mathbf{k}}^{(0)}\rangle$$

M. Royo and M. Stengel, Dynamical Response of Noncollinear Spin Systems at Constrained Magnetic Moments, Phys. Rev. X **16**, 011049 (2026)

$$\eta_{\alpha\beta\gamma}(\omega) = -\frac{4\pi}{\Omega} \text{Im} E_{\gamma}^{\mathcal{E}_{\alpha}\mathcal{E}_{\beta}}(\omega)$$

transparent regime: $\hbar\omega < \text{band-gap}$

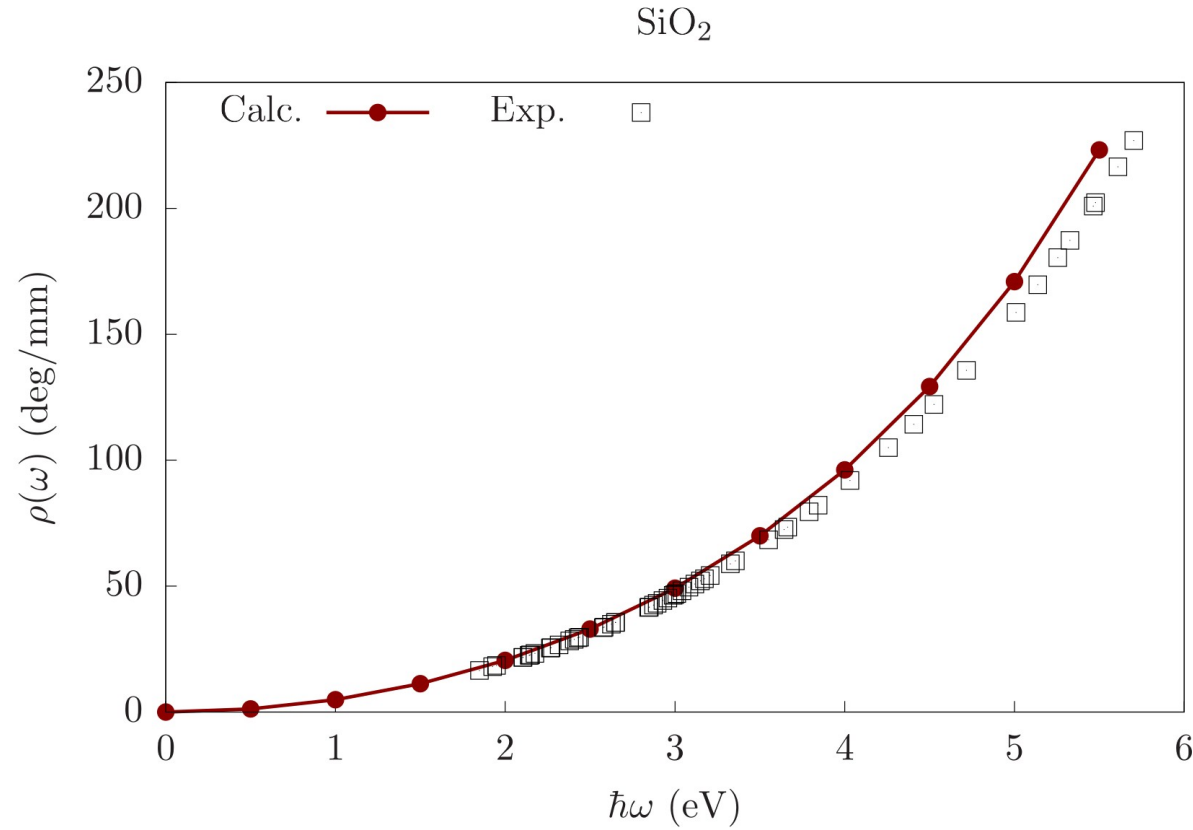
$$E_{\gamma}^{\mathcal{E}_{\alpha}\mathcal{E}_{\beta}}(\omega) = \int_{\Omega} \int n_{-\omega}^{\mathcal{E}_{\alpha}}(\mathbf{r}) \frac{\partial}{\partial q_{\gamma}} K_{\text{Hxc}}(\mathbf{r}, \mathbf{r}') \Big|_{\mathbf{q}=0} n_{\omega}^{\mathcal{E}_{\beta}}(\mathbf{r}') d^3r d^3r' + \int_{\text{BZ}} \Omega \frac{d^3k}{(2\pi)^3} \left[E_{\mathbf{k},\gamma}^{\mathcal{E}_{\alpha}\mathcal{E}_{\beta}}(\omega) - E_{\mathbf{k},\gamma}^{\mathcal{E}_{\alpha}\mathcal{E}_{\beta}*}(-\omega) \right]$$



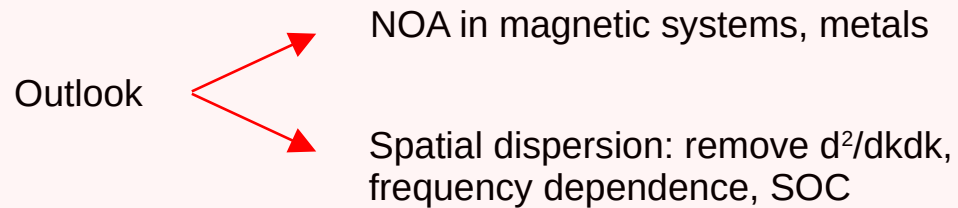
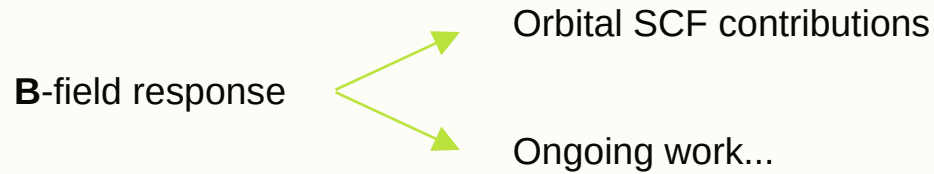
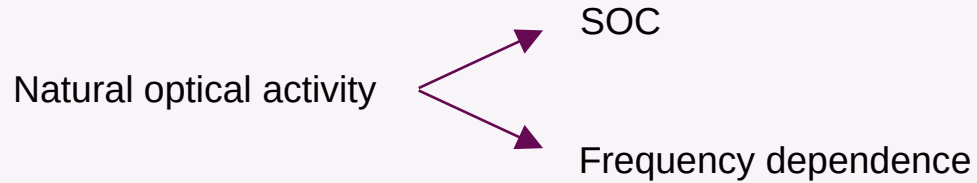
Consider both $+\omega$ and $-\omega$

$$n_{\omega}^{\mathcal{E}_{\beta}}(\mathbf{r}) = \int_{\text{BZ}} \Omega \frac{d^3k}{(2\pi)^3} \sum_m f_{m\mathbf{k}} \left(\langle u_{m\mathbf{k}}^{\mathcal{E}_{\beta}}(-\omega) | \mathbf{r} \rangle \langle \mathbf{r} | u_{m\mathbf{k}}^{(0)} \rangle + \langle u_{m\mathbf{k}}^{(0)} | \mathbf{r} \rangle \langle \mathbf{r} | u_{m\mathbf{k}}^{\mathcal{E}_{\beta}}(\omega) \rangle \right)$$

Test case: α -SiO₂



Summary and outlook



· Orb-**B**



rfmagn=4, ipert=2*natom+12

· Do we want spin+orb **B**?

rfmagn=5, ipert=natom+7